

CASE STUDY NO. 35 / SOFTWARE ENGINEERING & PROJECT MANAGEMENT

CARDIOWATCH

RHYTHM, PLAINLY.

Wearable Arrhythmia Detection with Regulatory Submission

Complete project report: six regulated-software deliverables, diagrams and evidence plan



THE SIX DELIVERABLES

- 01

Software Development Plan
- 02

SRS with Verification Matrix
- 03

Design Pack
- 04

Verification & Validation Plan
- 05

Estimation Report (COCOMO)
- 06

Configuration Management Plan
- MTTF 25,000 h / SENSITIVITY ≥ 96% / SPECIFICITY ≥ 98% / 18 KLOC / EAF 1.40

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PRODUCT PROMISE

Record. Check.
Understand.

Alerts stay
on device.

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SECTION 00

Project overview and UI prototype

The problem, the given data, the product idea, and the website prototype that makes it concrete.

0.1 The problem in plain words

A medical-device company is building **CardioWatch**, a wrist wearable that looks for **atrial fibrillation (AF)**, an irregular heart rhythm, and advises the wearer to seek care. Because it is a regulated medical device, the software must be built under a documented lifecycle, every requirement must be traced to verification evidence, and every design decision must be justified in a submission file. The engineering team is used to shipping weekly and sees this documentation as a burden. This report shows what the documentation is **for**, and produces all of it.

THE TWO WAYS TO GET IT WRONG

False negative (device misses real AF): a person with AF is told nothing, so a diagnosis is missed.

False positive (device flags a healthy person): anxious healthy people flood cardiology clinics.

Every number and every document in this report exists to control one of these two errors.

0.2 Given project data (used exactly as supplied)

Item	Given figure	Used in
Reliability	MTTF target 25,000 hours for the detection firmware	REL-01; Section 4.4 (how to demonstrate it)
Accuracy	Sensitivity $\geq 96\%$, specificity $\geq 98\%$	ACC-01..03; Section 2.6 (false positives per 100,000); Section 4.3
Size and mode	18 KLOC, embedded mode	Section 5 (COCOMO)
Cost driver	Intermediate COCOMO with EAF = 1.40 (reliability + documentation)	Section 5.2 and 5.4

0.3 The CardioWatch website UI (frontend prototype)

A frontend prototype website was built so the product is concrete before the formal documents are written. It deliberately follows the same three-move flow the requirements describe: **Record** a guided ECG, **Check** the signal quality first, then **Understand** the result with direct next-step guidance. It also shows the safety idea that drives the architecture: **alerts stay on the device**. The prototype contains no live patient data and makes no diagnosis.

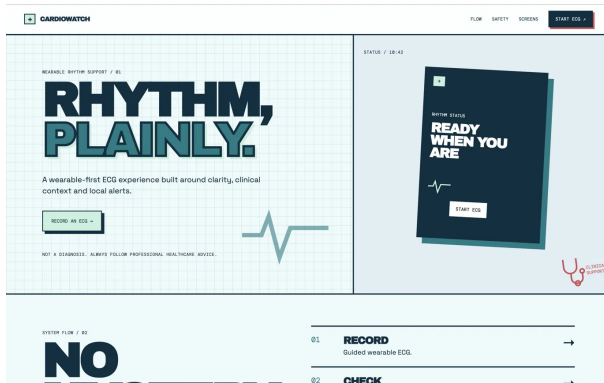


Fig 0.1 Landing page

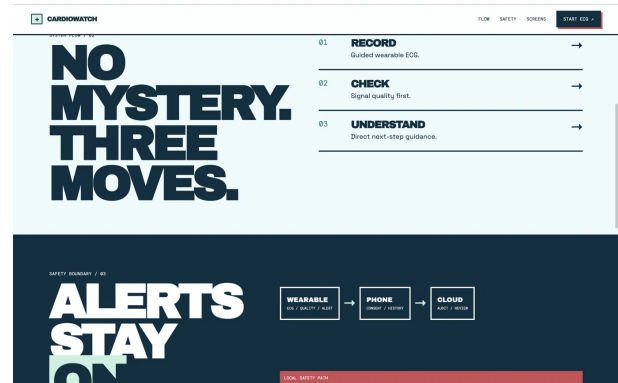


Fig 0.2 Record, Check, Understand + safety boundary

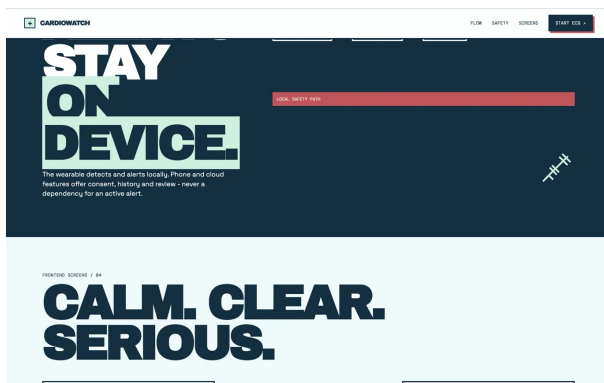


Fig 0.3 "Alerts stay on device"

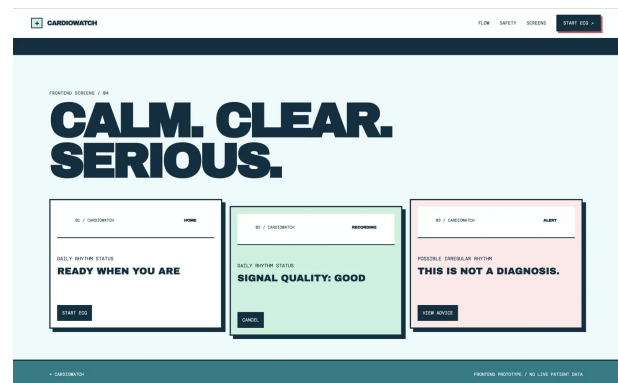


Fig 0.4 Home, Recording and Alert screens

How the UI supports the requirements

UI element	What it means for the system	Requirement
Home: "Ready when you are", START ECG	IDLE state; wearer starts a guided recording	FUN-01
Recording: "Signal quality: good", CANCEL	Quality checked before analysis; wearer can stop	FUN-02, 03, 05
Alert: "Possible irregular rhythm / This is not a diagnosis."	Mandatory labelling; cautious wording limits false-alarm harm	LAB-01
VIEW ADVICE	Next-step guidance to seek professional care	LAB-02
"Alerts stay on device"; Wearable → Phone → Cloud	Safety path is local; phone and cloud never needed for an alert	FUN-04, REL-03

DELIVERABLE 01

Software Development Plan

Lifecycle, phases, reviews and the V-Model verification mapping, with a plain explanation for a weekly-shipping team.

1.1 Purpose and scope

This plan states **how** the CardioWatch software will be developed: the lifecycle model, the phases, who does what, which reviews decide whether work may move on, and how every development phase is matched with a verification activity. It covers the wearable detection firmware (18 KLOC), the phone companion app and the cloud services, with the safety effort concentrated on the wearable.

1.2 Why the documentation exists (message to the weekly-shipping team)

A regulator cannot test every line of code. What it can do is inspect **evidence** that the product was built in a controlled way. The paperwork is not a tax on engineering; it is the proof that engineering was done properly. An auditor effectively asks three questions:

THE THREE AUDITOR QUESTIONS

1. **What did you intend?** (intended use, requirements, hazards)
2. **What did you build?** (design, code, baselines)
3. **How do you know it works?** (verification and validation evidence, linked back to each requirement)

The team does not have to stop shipping fast. It has to **separate fast internal builds from controlled releases**:

Weekly-shipping habit	What stays	What changes for a regulated device
Ship every week	Weekly internal builds and demos on test benches	Only a released baseline reaches a patient. Each release is frozen, verified and has release notes.
Merge when it works	Short branches, small changes, code review	Every change links to a requirement or hazard, is reviewed, passes regression tests and is recorded.
CI is green	Continuous integration stays	CI results are archived against the baseline ID and become verification evidence.
Write docs afterwards	Lightweight notes while exploring	Requirements and design are approved before coding; documents are controlled records.
Hotfix in a day	A fast path for urgent problems	An expedited change-control path: quicker, but impact analysis and regression are never skipped.

1.3 What regulation actually requires

Regulation does **not** say "use waterfall" or "do not use agile". It requires a planned, documented, traceable and controlled process whose results can be inspected:

Area	Commonly used standard	In plain words	Where in this report
Software lifecycle	IEC 62304	Plan the work, define phases, keep records of each activity	Section 1
Risk management	ISO 14971	Find hazards (missed AF, false alarm), control them, prove the controls work	SRS 2.2, V&V 4
Quality system	ISO 13485	Controlled procedures, records, audits, corrective action	Section 6
Usability and labelling	IEC 62366-1 and labelling rules	The wearer must understand the screens, especially "not a diagnosis"	LAB requirements

Area	Commonly used standard	In plain words	Where in this report
Regulatory submission	Market route (e.g. FDA premarket, EU MDR technical file)	One file linking intent, design, verification and risk	Section 6.6

The exact submission route depends on the target market and is decided by Regulatory Affairs; the standards above are those most commonly applied.

1.4 Lifecycle choice: the V-Model (justified)

The **V-Model** is chosen because every development phase on the left has a matching verification or validation activity on the right. That one-to-one pairing is exactly what a regulator wants to see and what makes traceability natural.

- **Safety-critical:** a missed AF or a false alarm has real consequences, so test planning must start when requirements are written, not after coding.
- **Requirements are stable and externally driven:** accuracy, reliability and labelling targets are fixed by the project data and regulation.
- **Evidence by design:** each box produces a signed document, which becomes part of the submission file.
- **Early defect detection:** reviewing requirements and design against their later tests finds errors while they are cheap.
- **Auditor familiarity:** the V-Model maps directly to IEC 62304 activities.

How weekly shipping survives: inside each left-hand phase the team may iterate in short cycles and cut weekly internal builds. What is not allowed is sending an unverified build to a patient: only a tagged, signed, baselined release can be flashed to patient-facing devices.

1.5 V-Model and the verification counterpart of every phase

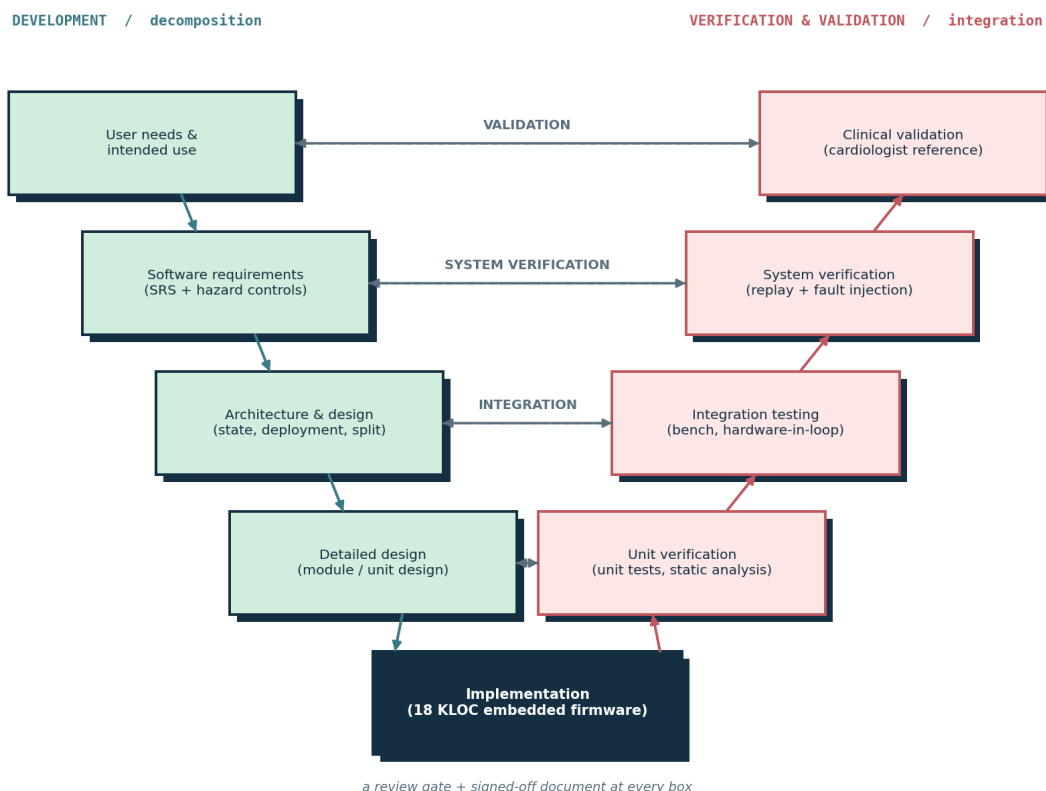


Figure 1.1 The V-Model for CardioWatch. Dashed lines pair each development phase with its verification or validation partner.

Development phase (left)	Output document	Verification counterpart (right)	Review gate	Evidence produced
User needs and intended use	Intended-use statement, hazard list	Clinical validation	Intended-use review (IUR)	Clinical validation report
Software requirements	SRS + verification matrix	System verification (replay, fault injection)	Requirements review (SRR)	System test report, replay report

Development phase (left)	Output document	Verification counterpart (right)	Review gate	Evidence produced
Architecture and design	Design pack, interface specs	Integration testing (bench, hardware-in-loop)	Design review (DR)	Integration test report
Detailed design	Module and unit specs	Unit verification (unit tests, static analysis)	Detailed design review	Unit results, analysis reports
Implementation	Source code, build scripts	Code review and inspection	Code review (each change)	Review records

1.6 Review gates

Gate	When	Who attends	Passes when
IUR: Intended-use review	End of needs phase	PM, clinical advisor, regulatory affairs	Intended use and hazards agreed; BL-0/BL-1 inputs ready
SRR: Requirements review	SRS complete	PM, SW lead, V&V lead, QA	Every requirement verifiable, has a method and hazard link
DR: Design review	Design pack complete	SW lead, V&V lead, QA, clinical advisor	State and deployment design approved; design decisions justified
TRR: Test readiness review	Before system verification	V&V lead, QA, CM manager	Test specs approved; build frozen as BL-3; tools controlled
RR: Release review	Before release / submission	All roles + regulatory affairs	All requirements verified, open defects acceptable, labelling matches reports

1.7 Safety classification

Software safety class under IEC 62304 is confirmed through the ISO 14971 risk analysis. Because a missed detection can contribute to serious harm, CardioWatch detection and alert software is **planned conservatively as Class C**, the strictest class. This means the full set of activities is applied: documented detailed design, unit-level verification, and complete traceability. Lower-risk parts (for example the history screen) may be justified at a lower class during the risk analysis, but the plan does not rely on that.

DELIVERABLE 02

SRS with Verification Matrix

Requirements written in verifiable form: every requirement has a verification method, an evidence type and an acceptance criterion.

2.1 Introduction

Intended use (draft). CardioWatch is a wrist-worn device that records a single-lead ECG on request, identifies patterns suggestive of atrial fibrillation in the intended adult population defined in the labelling, and advises the wearer to seek professional care. It **does not diagnose** and does not replace clinical judgement.

Key terms. **Sensitivity** = share of people with AF that the device flags. **Specificity** = share of people without AF left unflagged. **MTTF** = mean time to failure. **Verification methods:** **T** Test, **A** Analysis, **I** Inspection, **D** Demonstration.

2.2 Hazards that drive the requirements

ID	Hazard	Possible harm	Controlled by
H-01	Missed AF (false negative)	Undetected AF, delayed treatment	ACC-01, ACC-04, FUN-04, REL-02
H-02	False alarm (false positive)	Anxiety, unnecessary clinic visits	ACC-02, ACC-03, FUN-02, FUN-03, LAB-01
H-03	False reassurance	Wearer ignores real symptoms after a "normal" result	LAB-03, FUN-05
H-04	Silent device failure	Wearer believes monitoring is working	REL-01, REL-02
H-05	Alert delayed or suppressed by connectivity	Late care	FUN-04, FUN-06, REL-03
H-06	Misleading or missing labelling	Result misunderstood	LAB-01..LAB-05
H-07	Unverified or tampered firmware	Unpredictable behaviour	SEC-01

2.3 What a verifiable requirement looks like

Weak (not verifiable)	Strong (verifiable)
"The device shall be accurate."	"The detection shall achieve sensitivity $\geq 96\%$ on a locked validation set."
"The system shall be reliable."	"The detection firmware shall have an MTTF of at least 25,000 hours."
"Alerts shall be clear."	"The alert screen shall show POSSIBLE IRREGULAR RHYTHM and THIS IS NOT A DIAGNOSIS."

2.4 Requirements and verification matrix

Each requirement is one verifiable statement with its verification method, evidence type and acceptance criterion. The hazard it controls is listed in 2.2.

ID	Requirement	Method	Evidence type	Acceptance criteria
FUN-01	When the wearer selects START ECG, the device shall record a guided single-lead ECG and show recording status continuously until the recording completes or the wearer selects CANCEL.	T, D	Bench test log + demonstration record	Every step of test script TS-FUN-01 passes on all executed runs; status is never blank during recording.
FUN-02	Before analysis, the device shall assign every recording a signal-quality result (good or poor) and shall not issue a possible-AF alert from a recording rated poor.	T	Replay report (poor-quality subset)	Zero alerts from recordings labelled poor-quality in the locked replay set; SQI agreement with annotators reported.

ID	Requirement	Method	Evidence type	Acceptance criteria
FUN-03	The recording screen shall show SIGNAL QUALITY: GOOD when quality is good and shall give corrective guidance when it is poor.	I, D	UI inspection record + demonstration	Both messages appear in the reviewed screens, wording matches the approved labelling text.
FUN-04	When AF is detected, the device shall raise the alert on the wearable itself without needing the phone or the cloud.	T	Fault-injection log (Bluetooth off, no network)	With phone and cloud unavailable, 100% of known-AF test signals still produce the on-device alert.
FUN-05	On CANCEL the device shall stop recording, discard the partial recording and return to HOME without showing any result.	T	Bench test log	All CANCEL test cases end in HOME with no result displayed and partial data discarded.
FUN-06	The device shall store every result locally with date, time and firmware version, and shall sync stored results to the phone when connected.	T	Integration test log	No stored result is lost across the forced disconnect and reconnect test set; version recorded on each result.
ACC-01	The AF detection shall achieve sensitivity $\geq 96\%$.	T	Replay report + clinical validation report	Accept if at most 6 of at least 300 AF recordings are missed (one-sided 95% confidence that sensitivity $\geq 96\%$, Sec 4.3).
ACC-02	The AF detection shall achieve specificity $\geq 98\%$.	T	Replay report + clinical validation report	Accept if at most 6 of at least 600 non-AF recordings are falsely flagged (one-sided 95% confidence that specificity $\geq 98\%$).
ACC-03	The expected false positives shall not exceed 2,000 per 100,000 healthy wearers (derived from ACC-02).	A	Analysis report (Sec 2.6)	Calculation reviewed; observed false-positive rate $\times 100,000 \leq 2,000$.
ACC-04	Accuracy shall be measured only on data not used to train or tune the algorithm.	I	Data-lineage record (dataset hashes)	Zero overlap between tuning and validation datasets shown by hashes and records.
REL-01	The detection firmware shall have an MTTF of at least 25,000 hours.	A, T	Reliability report (analysis + zero-failure test + field plan)	Zero failures in at least 75,000 device-hours (95% confidence), or an equivalent evaluation allowing failures per Sec 4.4.
REL-02	On sensor failure, memory error or watchdog reset the device shall enter the FAULT state, show "detection unavailable" and shall never display a normal result.	T	Fault-injection log	Every injected fault ends in FAULT with the message shown; zero normal results displayed.
REL-03	A phone or cloud outage, or a lost Bluetooth link, shall not delay or suppress an on-device alert.	T	Fault-injection log	Alert behaviour is identical with the links up or down.
SEC-01	The wearable shall run only firmware that is signed and listed in the released baseline.	T, I	Signature-check test + baseline record	Unsigned or modified images are rejected on every attempt.
LAB-01	The alert screen shall show POSSIBLE IRREGULAR RHYTHM and the statement THIS IS NOT A DIAGNOSIS.	I	Labelling review record	Both texts present, exact wording per approved labelling specification.
LAB-02	The alert screen shall offer VIEW ADVICE, which shall advise the wearer to seek professional care.	D, I	Demonstration + clinical wording review	Advice reachable in one tap; text approved by the clinical advisor.
LAB-03	A no-irregularity result shall state that it does not rule out other conditions.	I	Labelling review record	Statement present on every normal-result screen variant.
LAB-04	The instructions for use and About screen shall state intended use, limitations and the verified accuracy figures.	I	Labelling-versus-report cross-check	Every stated figure matches the signed V&V report.
LAB-05	The About screen shall show the software version, which shall map to a baseline ID.	I	CM record check	Version on every release build maps to a baseline record.

2.5 Verification coverage

TRACEABILITY STATUS

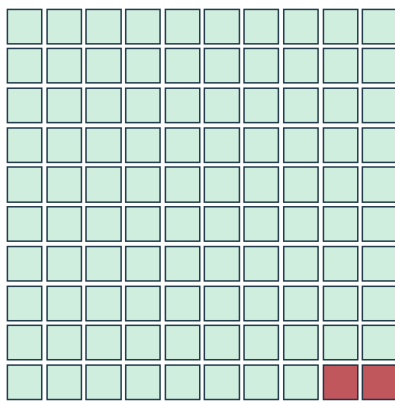
19 requirements, 19 with a verification method and acceptance criterion (100%). Test: 11, Inspection: 8, Demonstration: 3, Analysis: 2. Every requirement controls at least one hazard and every hazard has at least one requirement.

2.6 Accuracy in patient numbers: false positives and their clinical consequence

Specificity $\geq 98\%$ means the false-positive rate is at most $100\% - 98\% = 2\%$. For every **100,000 healthy wearers** the expected number wrongly flagged is:

$$100,000 \times 0.02 = 2,000 \text{ false positives}$$

100,000 healthy wearers (1 square = 1,000 people)



Specificity $\geq 98\%$

at most 2% of healthy people may be flagged

$$100,000 \times 2\% = 2,000$$

false alarms per 100,000 healthy wearers

$$98 \text{ mint squares} = 98,000$$

correctly told "no irregularity"

$$2 \text{ coral squares} = 2,000$$

anxious people who may visit a clinic

Figure 2.1 What 98% specificity means for 100,000 healthy wearers: up to 2,000 receive a false alarm.

Healthy wearers	Maximum false alarms (2%)	Correctly left unflagged (98%)
10,000	200	9,800
100,000	2,000	98,000
1,000,000	20,000	980,000

Missed cases (sensitivity). Sensitivity $\geq 96\%$ means up to 4% of people with AF may be missed: at the limit, **4 in every 100 people with AF** (400 in 10,000) would receive no alert on a given recording.

Clinical consequence, in patient numbers

- At the limit, **2,000 of every 100,000 healthy wearers** are told "possible irregular rhythm". Some will book a cardiology or GP appointment, consuming clinic time that real AF patients need.
- At one million wearers the same rate means **20,000 anxious people** with no AF. The harm is worry, repeat ECGs and unnecessary visits, not physical injury.
- The 98% figure is a rate **per recording**. A wearer who records often has repeated chances of a false alarm, so real-world alarms per person can exceed the per-recording rate.
- Raising sensitivity (fewer missed AF cases) usually raises false positives. The balance is set on tuning data only and then **locked** before validation (ACC-04).

MITIGATIONS

Design controls that reduce harm from false positives: (1) a signal-quality gate so noisy recordings never trigger an alert (FUN-02); (2) the wording "possible irregular rhythm" and "this is not a diagnosis" (LAB-01); (3) advice that is calm and specific (LAB-02); (4) clinic-load monitoring after release feeding the post-market process (Section 6).

DELIVERABLE 03

Design Pack

Architecture, state diagram, deployment diagram, and a defended on-device versus cloud split.

3.1 Design principles

- **The safety path is local.** Capture, quality check, detection and alert all run on the wearable.
- **Fail safe.** When the device is unsure or faulty it says so; it never shows a normal result by default (REL-02).
- **Optional outer tiers.** Phone and cloud add convenience and oversight but are never needed for an alert (REL-03).
- **Simple, finite states** so every behaviour can be tested and traced.
- **Everything traceable** to a requirement and a hazard.

3.2 Logical architecture

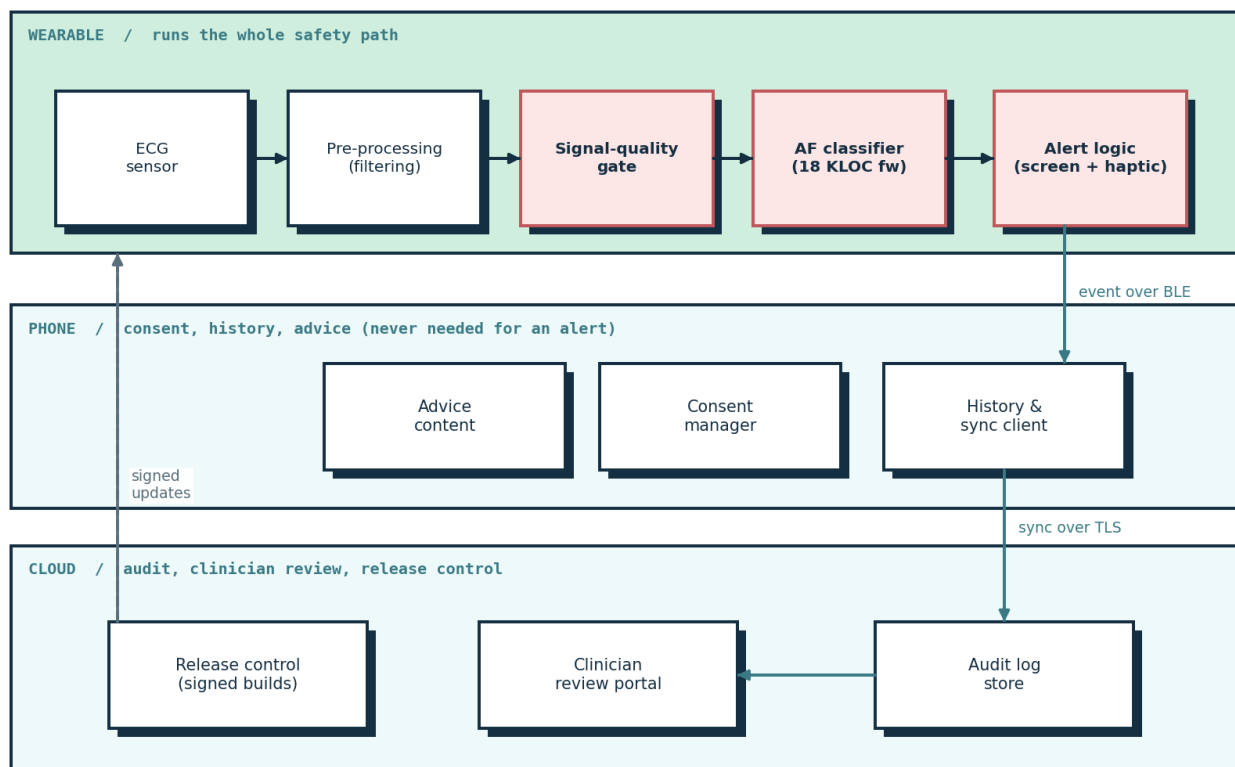


Figure 3.1 Logical architecture. Coral blocks form the safety path: quality gate, AF classifier and alert logic, all on the wearable.

3.3 State diagram: detection and alerting

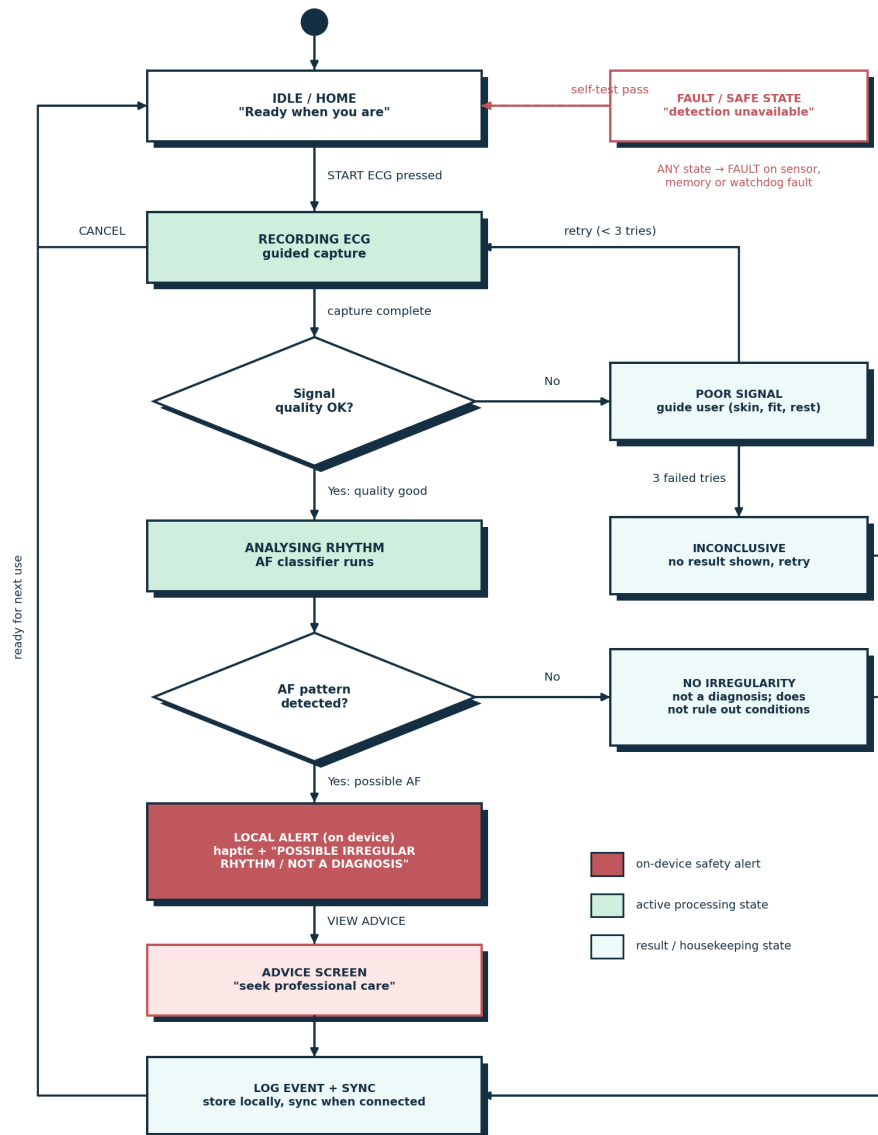


Figure 3.2 State diagram for detection and alerting. The coral state is the on-device safety alert; FAULT can be entered from any state.

Reading the diagram: the wearer starts a recording, the signal is checked, and only good signals are analysed. A possible-AF result raises the coral on-device alert; everything is logged and synced later. Any fault sends the device to a safe FAULT state, never to a silent "normal".

3.4 Deployment diagram: wearable, phone and cloud

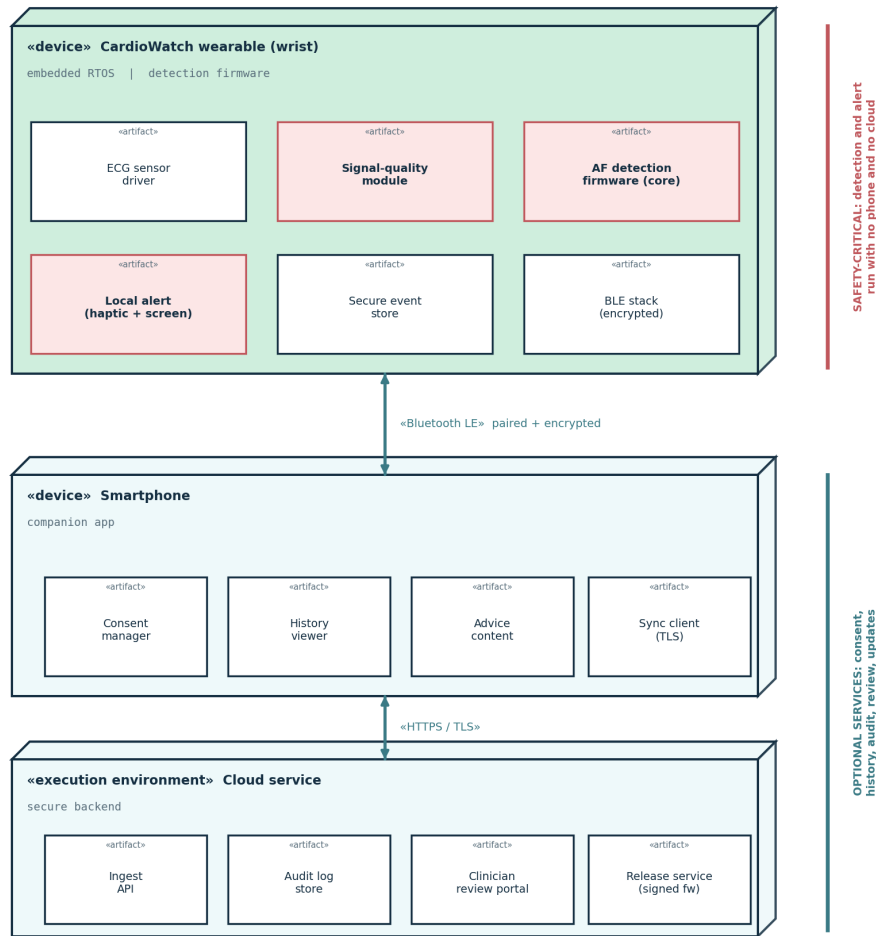


Figure 3.3 UML deployment diagram. The safety-critical artifacts (coral) live on the wearable; phone and cloud are optional services.

Links: Bluetooth LE (paired, encrypted) carries events and signed updates; HTTPS/TLS carries consent, history and audit data. If either link fails, alerts are unaffected and data queues for later sync.

3.5 On-device versus cloud split, and why

Function	Runs on	Why there
ECG capture and filtering	Wearable	Needs the raw signal with exact timing; streaming raw ECG would drain the battery
Signal-quality check	Wearable	Must run before analysis and give instant feedback to the wearer
AF detection	Wearable	Safety path: works offline, low delay, raw ECG need not leave the device (privacy)
Alert (haptic + screen)	Wearable	The wearer is at the wrist; the alert cannot wait for a network
Local event log	Wearable	A result must be recorded even when nothing is connected
Consent capture	Phone	Large screen for legal text; wearer can review and withdraw consent
History and trends	Phone	Storage and readable charts; not safety-critical
Advice content	Phone	Approved text can be updated under change control without touching detection firmware
Audit log, clinician review	Cloud	Long-term storage, access control, expert review
Release management	Cloud	Signed, controlled distribution; the wearable only accepts signed baseline firmware

JUSTIFICATION

Why not everything in the cloud? A network delay or outage would delay or suppress an alert (hazard H-05), the battery cost of streaming ECG is high, and raw ECG would leave the device. **Why not everything on the device?** Long-term history, expert review, audit and controlled update distribution need storage, access control and a larger display.

DELIVERABLE 04

Verification & Validation Plan

Bench testing, recorded-signal replay and clinical validation, with sample sizes, and how MTTF is demonstrated without a three-year test.

4.1 Strategy: three layers of evidence

Verification asks "did we build the product right?" (does it meet the requirements). **Validation** asks "did we build the right product?" (does it work for real people). CardioWatch uses three layers, from many cheap tests to few realistic ones:

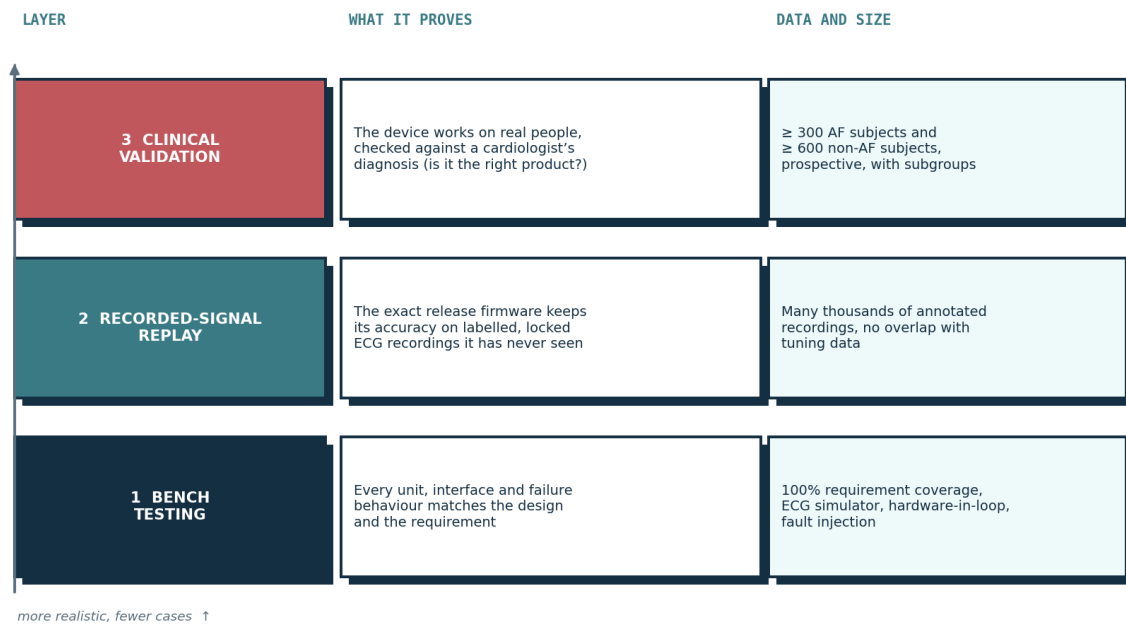


Figure 4.1 Verification and validation layers. Each layer proves something the layer below cannot.

4.2 Bench testing and recorded-signal replay

- **Bench:** unit tests, static analysis and peer review of the alert path; hardware-in-loop with an ECG simulator; every state transition in Figure 3.2; fault injection (sensor, memory, watchdog, Bluetooth off, phone and cloud down); unsigned-firmware rejection.
- **Replay:** the exact release firmware runs on bench hardware against cardiologist-annotated ECG recordings (AF, normal, other arrhythmias, noisy), far faster than real time. Candidate sources: public annotated ECG databases (e.g. PhysioNet challenge sets) plus company recordings.
- **Independence:** the replay corpus is locked and hashed; no recording used to tune thresholds may appear in it (ACC-04).
- **Outputs:** confusion matrix, sensitivity and specificity with confidence bounds, and the count of alerts from poor-quality recordings (must be zero, FUN-02).
- **Limit:** replay proves the algorithm, not the sensor on real wrists, so clinical validation is still needed.

4.3 Clinical validation and sample sizes

A prospective study compares CardioWatch against a **cardiologist-adjudicated reference diagnosis** in real wearers. Sample sizes come from a simple binomial argument: decide in advance how many misses are acceptable, so that passing the test gives 95% confidence that the requirement is truly met.

	Sensitivity (ACC-01)	Specificity (ACC-02)
Requirement	$\geq 96\%$ (miss rate $\leq 4\%$)	$\geq 98\%$ (false-positive rate $\leq 2\%$)

	Sensitivity (ACC-01)	Specificity (ACC-02)
Subjects needed	≥ 300 AF subjects	≥ 600 non-AF subjects
Accept if	at most 6 misses	at most 6 false positives
Why this is safe	If the true miss rate were 4%, seeing 6 or fewer misses in 300 has probability 4.3% (below 5%)	If the true false-positive rate were 2%, seeing 6 or fewer in 600 has probability 4.4% (below 5%)
Chance a good device passes	Planning assumption: true sensitivity 98.5% gives 83% chance to pass	Planning assumption: true specificity 99.2% gives 79% chance to pass

The 98.5% and 99.2% design margins are planning assumptions for sizing the study, not claims; the requirements remain the given 96% and 98%.

- **Subgroups:** recruit across age, sex, skin tone, wrist size and activity so that bias is visible; subgroup targets are fixed in the protocol.
- **Conduct:** ethics approval, informed consent, locked analysis plan, blinded adjudication of the reference diagnosis.
- **Pass rule:** both ACC-01 and ACC-02 must pass on the locked build; any algorithm change restarts the affected evidence (Section 6).

4.4 Demonstrating MTTF = 25,000 hours without a 25,000-hour test

25,000 hours is about **2.85 years** of continuous operation. A single device cannot be tested that long before launch, and it is not necessary. MTTF is shown by an **argument supported by evidence**, built from three independent legs:

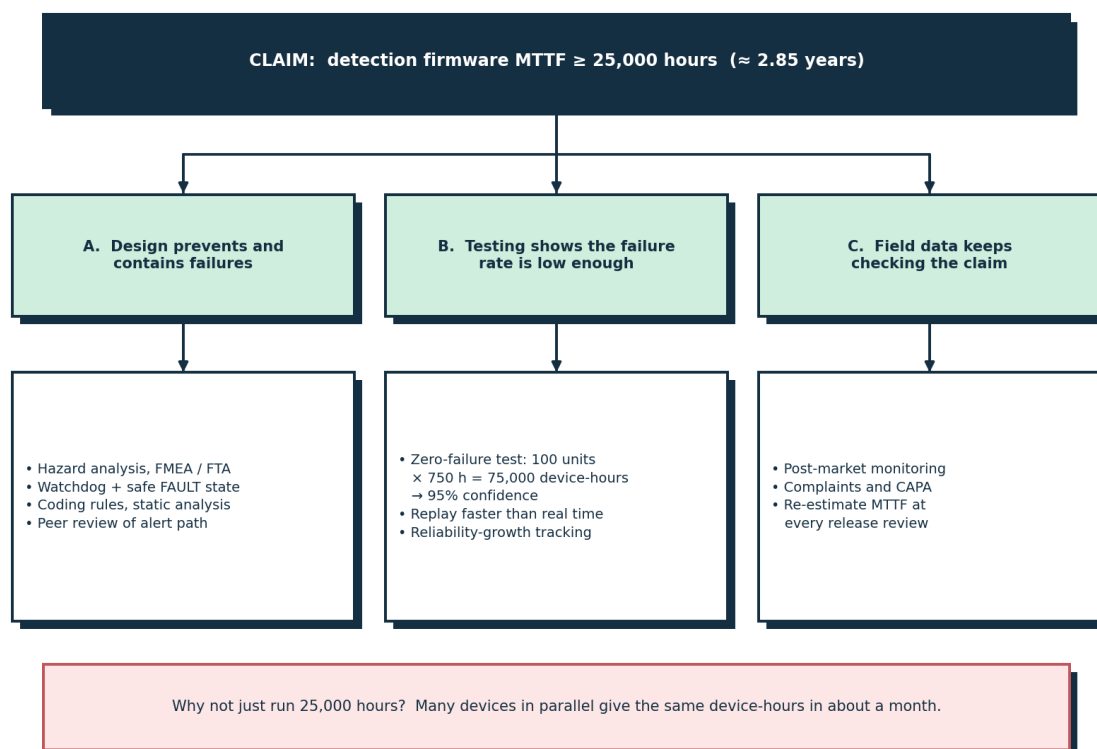


Figure 4.2 Reliability argument: design, test and field evidence together support the MTTF claim.

The statistics behind the test leg

Run many devices in parallel and add up their hours. If the failure rate is constant (exponential model) and **no failures** occur, the device-hours needed for confidence C are:

$$T = \text{MTTF} \times \ln(1 / (1 - C))$$

Confidence	Device-hours (0 failures)	Example test plan
90%	$25,000 \times \ln(10) = 57,565$	100 units × 576 h (about 24 days)
95%	$25,000 \times \ln(20) = 74,893$	100 units × 750 h (about 31 days)

If failures occur the test can continue, but more device-hours are needed (for example 118,597 device-hours at 95% confidence if one failure is allowed).

WHAT THE ARGUMENT DOES NOT CLAIM

Honest limits. Replay at faster than real time speeds up failures caused by *inputs* (software logic errors). Failures that depend on *elapsed time* (memory leaks, battery, wear) need real-time soak testing, which is why the 100-unit parallel test runs in real time. Design evidence (FMEA, watchdog, coding rules) and field data cover what testing cannot, and the MTTF estimate is re-checked at every release review.

4.5 Entry and exit criteria, defects and records

- **Entry to system verification:** test specs approved, build frozen as BL-3, tools and data under control.
- **Exit:** all requirements verified, matrix 100% complete, replay and fault-injection results approved.
- **Defects:** no critical defect (missed or wrong alert, normal shown in FAULT) at release; major defects only with an approved risk justification.
- **Every test run records** baseline ID, test spec version, tool versions, dataset hash, tester, date, result and defect IDs.

DELIVERABLE 05

Estimation Report

Basic and Intermediate COCOMO for 18 KLOC in embedded mode, with the EAF 1.40 multiplier explained.

5.1 Inputs and method

The case study fixes **size = 18 KLOC**, **mode = embedded** (tight hardware, regulatory and timing constraints) and, for Intermediate COCOMO, **EAF = 1.40** for required reliability and documentation. COCOMO estimates development effort E (person-months), duration D (months) and average staffing.

Model	Effort equation	Duration equation	Embedded constants
Basic	$E = a \times (\text{KLOC})^b$	$D = c \times E^d$	$a = 3.6, b = 1.20, c = 2.5, d = 0.32$
Intermediate	$E = a \times (\text{KLOC})^b \times \text{EAF}$	$D = c \times E^d$	$a = 2.8, b = 1.20, c = 2.5, d = 0.32$

5.2 COCOMO calculations

Step	Basic COCOMO	Intermediate COCOMO (EAF 1.40)
1 Size power	$18^{1.20} = 32.09$	$18^{1.20} = 32.09$
2 Effort	$E = 3.6 \times 32.09 = 115.5 \text{ PM}$	$E = 2.8 \times 32.09 = 89.8 \text{ nominal}; \times 1.40 = 125.8 \text{ PM}$
3 Duration	$D = 2.5 \times 115.5^{0.32} = 11.43 \text{ months}$	$D = 2.5 \times 125.8^{0.32} = 11.74 \text{ months}$
4 Average staff	$115.5 / 11.43 = 10.1 \text{ people}$	$125.8 / 11.74 = 10.7 \text{ people}$
5 Productivity	156 LOC per person-month	143 LOC per person-month

5.3 Comparison against the basic estimate

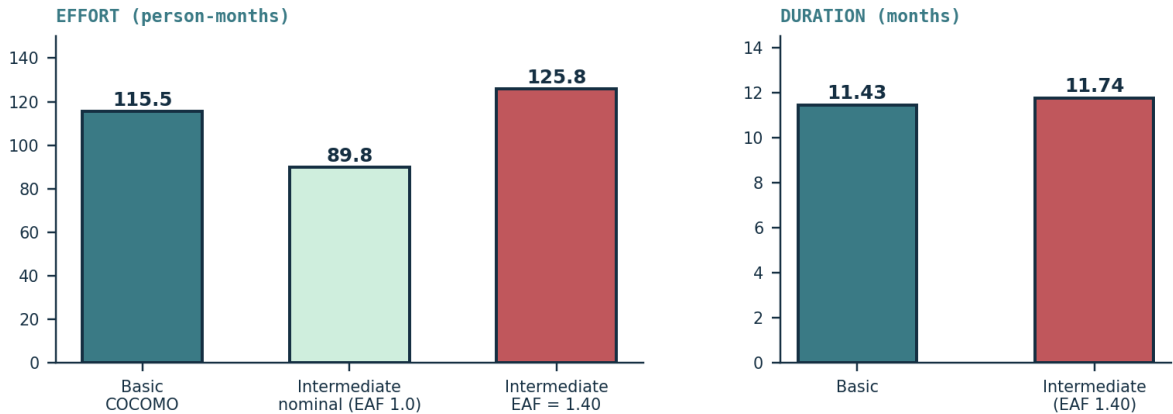


Figure 5.1 Effort and duration: Basic COCOMO, Intermediate nominal, and Intermediate with EAF = 1.40.

Measure	Basic	Intermediate (EAF 1.40)	Difference
Effort (person-months)	115.5	125.8	+10.3 (+8.9%)
Duration (months)	11.43	11.74	+0.32 (+2.8%)
Average staff	10.1	10.7	+0.6
Productivity (LOC/PM)	156	143	-13

The Intermediate estimate is about **9% more effort** than Basic but only about **3% longer**, because duration grows with the cube-root-like exponent 0.32: extra work is absorbed mostly by more people in parallel. Note that the Basic embedded estimate (115.5 PM) is already 29% above the Intermediate nominal (89.8 PM); the 1.40 multiplier pushes the plan further because CardioWatch’s reliability and documentation needs exceed an

average embedded project.

5.4 What the 1.40 multiplier physically represents

EAF = 1.40 means **40% more person-months** than the same code would need in an ordinary project: 35.9 extra person-months over the 11.7-month schedule, roughly **3 additional full-time engineers**. None of that extra effort writes more code. In a regulated project it is physically spent on:

- **Reviews and inspections** of requirements, design and code (gates SRR, DR, TRR, RR).
- **Independent verification:** writing test specs, building replay and fault-injection rigs, running and recording tests.
- **Traceability upkeep:** keeping hazards, requirements, design, code and tests linked and consistent.
- **Documentation as controlled records:** drafting, approving, e-signing and baselining each deliverable.
- **Rework driven by stricter exit criteria:** a defect in the alert path must be fixed, re-verified and re-documented.

PLAIN-LANGUAGE MEANING

In standard COCOMO the value 1.40 is also the rating for **very high required reliability**, where a failure risks human safety. That is exactly CardioWatch's situation: the multiplier is the price of producing **evidence**, not of typing code. It also answers the weekly-shipping team: the documentation burden is a measurable 40% of the effort, and it is what makes release possible.

DELIVERABLE 06

Configuration Management Plan

Baselines, change control and submission document control, designed to survive an auditor’s inspection.

6.1 Purpose and scope

Configuration management (CM) makes sure the team always knows **exactly what exists, which version is approved, what changed, why, and who approved it**. It covers documents, source code, third-party software, tools, data sets, binaries and the submission file. The **Change Control Board (CCB)** consists of the project manager (chair), software lead, V&V lead, QA / regulatory affairs and the clinical advisor.

6.2 Configuration items

Category	Items	Identification	Stored in
Documents	SDP, CM plan, SRS, design pack, V&V plan, test specs and reports, risk file, trace matrix, labelling	CW-<type>-<nnn>vX.Y	Document system with e-signature
Source code	Firmware modules, app code, build scripts, configuration	Repository path + commit ID	Git, protected main branch
Third-party software (SOUP)	RTOS, libraries: version and known-anomaly list	SOUP-nn	Repository + SOUP register
Tools	Compiler, static analyser, test frameworks, versions	TOOL-nn + container digest	Tool register
Data	Replay corpus, tuning set, validation set, thresholds	DATA-nn + hash	Read-only controlled storage
Binaries	Firmware images, signed	MAJOR.MINOR.PATCH+build	Artifact store
Submission file	Index and approved final documents	SUB-nn	Document system

IDENTIFICATION

Versioning: documents carry an ID and version (Draft → In review → Approved → Baselined → Superseded); software uses MAJOR.MINOR.PATCH+build and every release is a signed Git tag equal to a baseline ID. Weekly internal builds are labelled *development builds* and are never flashed to patient-facing devices. A baseline stores commit, tool versions and dataset hashes so the identical binary can be rebuilt.

6.3 Baselines

FROZEN SNAPSHOTS: after approval, nothing changes except through a change request

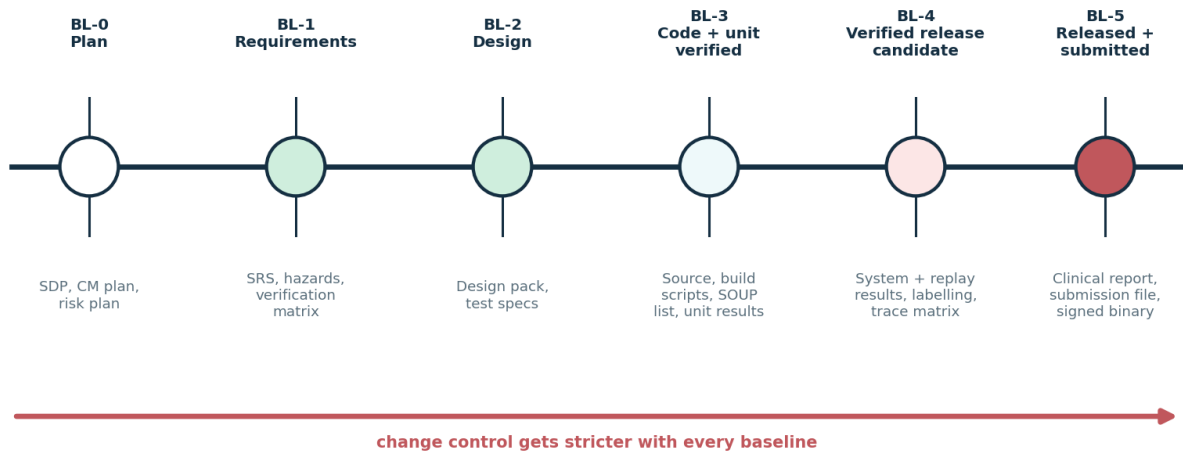


Figure 6.1 Baseline sequence BL-0 to BL-5. Each baseline is a frozen, approved snapshot.

Baseline	Frozen contents	Approved at	Entry criteria
BL-0 Plan	SDP, CM plan, risk plan	IUR	Plans reviewed and signed
BL-1 Requirements	SRS, hazard list, verification matrix	SRR	Every requirement verifiable, 100% matrix coverage
BL-2 Design	Design pack, interface specs, test specs	DR	Design decisions justified; trace to requirements complete
BL-3 Code + unit verified	Source, build scripts, SOUP list, unit results	TRR	Unit results pass; static analysis clean; reviews closed
BL-4 Release candidate	System, replay and fault-injection results, labelling, trace matrix	RR	All requirements verified; defects meet release rules
BL-5 Released / submitted	Clinical validation report, submission file, signed binary	RR + regulatory sign-off	Labelling matches V&V report; configuration audits passed

6.4 Change control

After a baseline is approved, **nothing in it changes except through a change request (CR)**. The same path applies to code and documents.

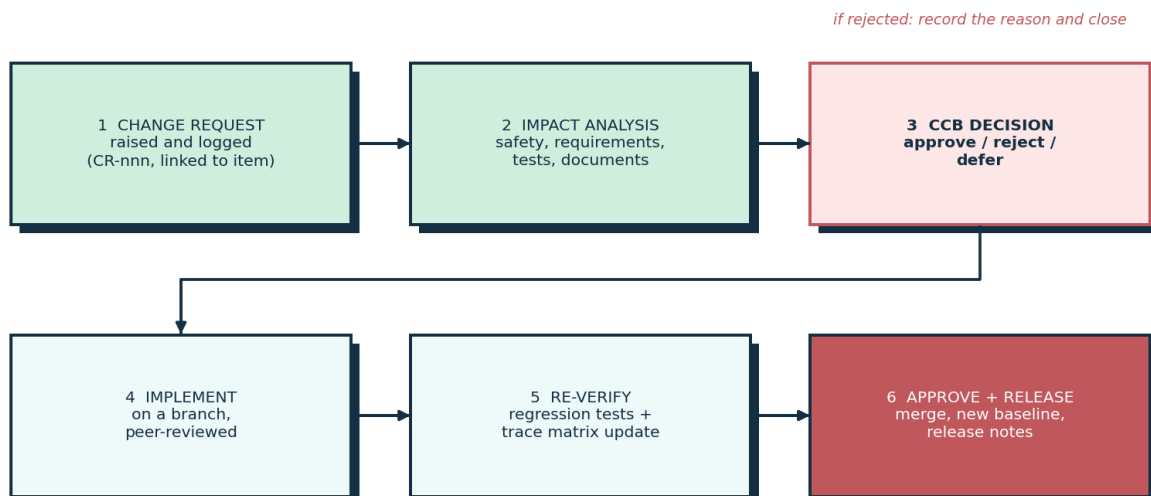


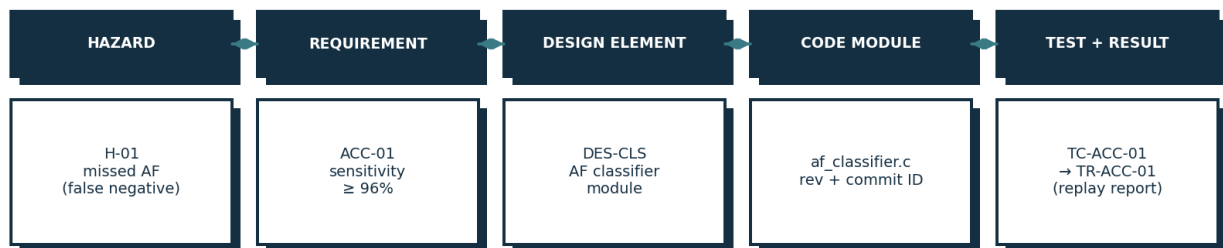
Figure 6.2 Change control flow from request to new baseline.

Class	Example	Path	Approval
A Document only	Typo, formatting	Edit, peer review, version bump	Document owner + reviewer
B Normal software change	Refactor outside the alert path	Full path in Figure 6.2	CCB
C Safety-affecting	Change to thresholds, alert logic, quality gate or labelling	Full path + risk file update + full regression + re-assessment of validation evidence	CCB with clinical advisor and regulatory check
D Emergency fix	Urgent field problem	Expedited path: CCB chair and V&V lead decide first, full record completed straight after; no step is skipped	CCB chair + V&V lead, ratified by CCB

Code control: protected main branch, short feature branches, a frozen release branch per baseline; a pull request needs a reviewer, green CI and a requirement or CR link; only a signed tag equal to a baseline ID can become a releasable binary (SEC-01).

6.5 Traceability scheme

every link is two-way: forward trace (is it built and tested?) and backward trace (why does it exist?)



illustrative example of one trace chain

Figure 6.3 Traceability chain from hazard to test result, navigable in both directions.

- **Stable IDs** (H-, REQ prefixes such as ACC-, DES-, TC-, TR-) are never reused.
- The trace matrix is **generated from the repository and document system**, not hand-maintained.
- A CI check reports **orphans**: a requirement with no test, a test with no requirement, code with no requirement, a hazard with no control. Orphans block a baseline.

6.6 Submission document control

The submission file is assembled **only** from approved, baselined documents. After submission it is read-only; later changes go through the change process and are submitted as a supplement or update as the market route requires.

- Document index lists every file, version, state and signatories; controlled copies carry the version and print date.
- Full audit trail (who, what, when, why) is retained for the product life plus the required retention period.

6.7 Surviving an auditor's inspection

Auditor question	Where the answer lives	How it is shown
What exactly shipped as version X?	Baseline BL-5 record + signed binary hash	Show the manifest and verify the hash
Why does this module exist?	Backward trace to requirement and hazard	Open the trace matrix at the module
Prove ACC-01 was verified	ACC-01 → TC-ACC-01 → TR-ACC-01	Open the replay and clinical reports
What changed since the last submission, and who approved it?	CR log, CCB minutes, diff between baselines	List CRs between BL-4 and BL-5
Can you rebuild the same binary?	Tool register, container digest, signed tag	Rebuild and compare hashes
Was the test data kept separate from tuning data?	DATA register hashes (ACC-04)	Show zero overlap

SECTION 07

Outcomes checklist, references and glossary

Confirmation that every required outcome and deliverable has been produced.

7.1 Required outcomes

Required outcome	Where delivered
A verification matrix where every requirement has an evidence type attached	Section 2.5 (19 of 19 requirements covered)
A false-positive calculation with its clinical consequence in patient numbers	Section 2.6: 2,000 per 100,000 healthy wearers; 20,000 per million
A defended on-device versus cloud split	Sections 3.4 and 3.5
An explanation of reliability demonstration that does not require a three-year test	Section 4.4: 75,000 device-hours in parallel (95% confidence) plus design and field evidence
A traceability and baseline scheme that would survive an auditor's inspection	Sections 6.3, 6.5, 6.7

7.2 Required deliverables

Deliverable	Section	Includes
Software Development Plan	1	Lifecycle, phases, reviews, V-Model verification mapping, regulation explained
SRS with Verification Matrix	2	Hazards, 19 verifiable requirements, method and acceptance criteria for each
Design Pack	3	Architecture, state diagram, deployment diagram, on-device/cloud rationale, decision log
Verification & Validation Plan	4	Bench, replay, clinical strategy with sample sizes; MTTF argument
Estimation Report	5	Basic (115.5 PM, 11.4 mo) and Intermediate (125.8 PM, 11.7 mo) COCOMO; 1.40 interpreted
Configuration Management Plan	6	Baselines, change control, submission document control, traceability

7.3 Conclusion

CardioWatch shows why a fast-moving team accepts structure when the product is regulated: the documents are the evidence that the two dangerous errors, a missed AF and a false alarm, are understood, bounded and verified. The V-Model pairs every phase with proof; the safety path stays on the wearable; the estimate (126 person-months over about 12 months) honestly includes the cost of that proof; and configuration management keeps every claim reproducible.

7.4 Standards and references

- IEC 62304: Medical device software, software life cycle processes.
- ISO 14971: Application of risk management to medical devices.
- ISO 13485: Medical devices, quality management systems.
- IEC 62366-1: Application of usability engineering to medical devices.
- B. W. Boehm, *Software Engineering Economics*, Prentice Hall, 1981 (Basic and Intermediate COCOMO).
- Course material: Software Engineering & Project Management, B.Tech CSE, ITM Skills University.